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Technical Specification

3rd Generation Partnership Project; Technical Specification Group Services and System Aspects; 5G Security Assurance Specification (SCAS) for the Network Repository Function (NRF) network product class (Release 20)

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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

x the first digit:

1 presented to TSG for information;

2 presented to TSG for approval;

3 or greater indicates TSG approved document under change control.

y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document. In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document contains objectives, requirements and test cases that are specific to the NRF network product class. It refers to the Catalogue of General Security Assurance Requirements and formulates specific adaptions of the requirements and test cases given there, as well as specifying requirements and test cases unique to the NRF network product class.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 33.117: "Catalogue of general security assurance requirements".

[3] 3GPP TS 33.501: "Security architecture and procedures for 5G system".

[4] 3GPP TS 23.502: "Procedures for the 5G System".

[5] 3GPP TS 29.510: "5G System; Network function repository services; Stage 3".

[6] 3GPP TR 33.926: "Security Assurance Specification (SCAS) threats and critical assets in 3GPP network product classes".

[7] 3GPP TS 23.501: "System Architecture for 5G System (5GS)".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

NF	Network Function
NRF	Network Repository Function

4 NRF-specific security requirements and related test cases

4.1 Introduction

NRF specific security requirements include both requirements derived from NRF-specific security functional requirements in relevant specifications as well as the security requirements introduced in the present document derived from the threats specific to NRF as described in TR 33.926 [6].

4.2 NRF-specific adaptations of security functional requirements and related test cases

4.2.1 Introduction

The present clause describes the security functional requirements and the corresponding test cases for NRF network product class. The proposed security requirements are classified in two groups:

- Security functional requirements derived from TS 33.501 [3] and detailed in clause 4.2.2.
- General security functional requirements which include requirements not already addressed in TS 33.501 [3] but whose support is also important to ensure that NRF conforms to a common security baseline detailed in clause 4.2.3.

4.2.2 Security functional requirements on the NRF deriving from 3GPP specifications and related test cases

4.2.2.1 Security functional requirements on the NRF deriving from 3GPP specifications – general approach

In addition to the requirements and test cases in TS 33.117 [2], clause 4.2.2, a NRF shall satisfy the following:

It is assumed for the purpose of the present SCAS that a NRF conforms to all mandatory security-related provisions pertaining to a NRF in:

- 3GPP TS 33.501 [3]: "Security architecture procedures for 5G system";
- other 3GPP specifications that make reference to TS 33.501 [3] or are referred to from TS 33.501 [3].

Security procedures pertaining to a NRF are typically embedded in NF discovery/registration/access token request procedures and are hence assumed to be tested together with them.

4.2.2.2 NF discovery procedure

4.2.2.2.1 NF discovery authorization based on expected NF profile

Requirement Name: NF discovery authorization for specific scopes

Requirement Reference: TS 33.501 [3], clause 13.3.1.3, TS 23.502 [4], clause 4.17.4, and TS 29.510 [5], clause 6.2.3.2.3.1.

Requirement Description:

NRF is expected to be able to ensure that NF Discovery and registration requests are authorized as specified in TS 33.501 [3], clause 5.9.2.1.

The NRF checks that the values of the authorization parameters in the NF (Service) Profile of an NF Service Producer allows an NF Service Consumer to discover the NF Service Producer. In the response message, the NRF only returns information of those NF Service Producer instances that the NF Service Consumer is authorized to discover, as specified in the TS 33.501 [3], clause 13.3.1.3.

The NRF authorizes the Nnrf_NFDiscovery_Request. Based on the profile of the expected NF/NF service and the type of the NF service consumer, the NRF determines whether the NF service consumer is allowed to discover the expected NF instance(s). If the expected NF instance(s) or NF service instance(s) are deployed in a certain network slice, NRF authorizes the discovery request according to the discovery configuration of the Network Slice, e.g. the expected NF instance(s) are only discoverable by the NF in the same network slice as specified in TS 23.502 [4], clause 4.17.4.

Based on operator's policies, a discovery request not including the requester's information necessary to validate the authorization parameters in NF Profiles can be rejected or accepted but with only returning in the discovery response NF Instances whose authorization parameters allow any NF Service Consumer to access their services. The authorization parameters in NF Profile are those used by NRF to determine whether a given NF Instance / NF Service Instance can be discovered by an NF Service Consumer in order to consume its offered services (e.g. "allowedNfTypes", "allowedNfDomains", etc.), as specified in TS 29.510 [5], clause 6.2.3.2.3.1, Note 12.

If included, the requester-snssais IE is expected to contain the list of S-NSSAI of the requester NF. The NRF is expected to use this to return only those NF profiles of NF Instances allowing to be discovered from the slice(s) identified by this IE, according to the "allowedNssais" list in the NF Profile and NF Service as specified in TS 29.510 [5], clause 6.2.3.2.3.1.

Threat References: TR 33.926 [6], clause H.2.2.1, No Authorization of NF discovery based on Authorization Parameters

Test Case:

Test Name: TC_DISC_AUTHORIZATION_ALLOWED_PARAMETER

Purpose:

Ensure that the NRF being tested does not authorize a discovery request from an NF service consumer instance that lacks the correct authorization provided in the request, based on the parameters prefixed with “allowed” (e.g., allowedNfTypes, allowedNfDomains, allowedNssais…) provided by the NF service producer profile.

Procedure and execution steps:

Pre-Conditions:

- Test environment with the NF1 and NF2, which may be simulated. The NF2 will attempt to discover NF1.
- The NRF documentation provides information on whether unauthorized requests are rejected or accepted, but only returns NF Instances in the discovery response whose service the NF service consumer is authorized to access. If this is configurable, the tester is required to test both options.
- If the NRF under test does not support parameters from the allowedList in the table below, the test steps regarding these parameters are not applicable.

Execution Steps

For all Test Case specific parameters defined in the table 4.2.2.2.1-1, the tester shall repeat the following execution steps.

Table 4.2.2.2.1-1 Test Case Specific Parameter Sets

Test Case	*parameter NF1*	*parameter NF2*	*allowedList (NF1)*	*requester-type (NF2)*
A	NfType NF1	NfType NF2	allowedNfTypes	requester-nf-type
B	PLMN NF1	PLMN NF2	allowedPlmns	requester-plmn-list
C	FQDN NF1	FQDN NF2	allowedNfDomains	requester-nf-instance-fqdn
D	SNPN NF1	SNPN NF2	allowedSnpns	requester-snpn-list
E	S-NSSAI NF1	S-NSSAI NF2	allowedNssais	requester-snssais
F	S-NSSAI NF1 and PLMN NF1	S-NSSAI NF2 and PLMN NF2	allowedPlmns	requester-plmn-specific-snssai-list

1. The tester configures NF1 with *parameter NF1* and NF2 with *parameter NF2*, where the two parameter values are different. The tester should select the mandatory and optional profile parameters for NF1 and NF2 such that they do not conflict with other authorization test cases in this section.

2. The tester configures NF1 to ensure that it is not accessible by NF2 by disallowing *parameter NF2* via the *allowedList parameter* in the profile NF1.

3. The tester triggers NF1 to register as a new NF instance via the NFManagement API at the NRF under test.

4. The tester triggers NF2 to send an Nnrf_NFDiscovery_Request message to the NRF under test with *target-nf-type* set to NfType NF1 and *requester-type parameter* set to the corresponding *parameter NF2*.

Expected Results:

If the NRF under test is configured to reject unauthorized requests, the NRF responds with a "403 Forbidden" status code, as specified in clause 5.3.2.2.2 of TS 29.510 [5].

If the NRF under test is configured to accept unauthorised requests, but only returns NF instances whose authorisation is accepted in the discovery response, the discovery response will not contain any information about the NF1.

Expected format of evidence:

Evidence suitable for the interface, e.g., evidence can be presented in the form of packet trace (pcap-file).

4.2.2.3 Authentication for Indirect Communication

4.2.2.3.1 Correct handling of client credentials assertion validation failure

Requirement Name: Correct handling of client credentials assertion validation failure

Requirement Reference: TS 33.501 [10], clause 13.3.8.3: TS 29.500 [21], clause 6.7.5

Requirement Description:

The verification of the Client credentials assertion is performed by the receiving node, i.e., NRF or NF Service Producer in the following way:

- It validates the signature of the JWS as described in RFC 7515 [16].
- It validates the timestamp (iat) and/or the expiration time (exp) as specified in RFC 7519 [17].
- If the receiving node is the NRF, the NRF validates the timestamp (iat) and the expiration time (exp).
- If the receiving node is the NF Service Producer, the NF service Producer validates the expiration time and it may validate the timestamp.
- It checks that the audience claim in the client credentials assertion matches its own type.

It verifies that the NF instance ID in the client credentials assertion matches the NF instance ID in the public key certificate used for signing the assertion.

Threat References: TR 33.926 [4], clause 6.3.4.1, Incorrect validation of client credentials assertion

NOTE 1: The following test case only applies if the NF under test implements verification of client credentials assertions.

NOTE 2: This test case is an extension to TC_CLIENT_CREDENTIALS_ASSERTION_VALIDATION in TS 33.117 [2] and only includes the NRF-specific test for the validity of the iat value.

If the verification of the CCA fails at the receiving entity (e.g. NRF or NF service producer), a "403 Forbidden" response is returned with the cause attribute set to "CCA_VERIFICATION_FAILURE", as stated in TS 29.500 [21], clause 6.7.5.

Test Case:

Test Name: TC_CLIENT_CREDENTIALS_ASSERTION_VALIDATION_NRF

Purpose:

Verify that the NF under test correctly handles client credentials assertion validation failure. This test case specifically verifies that the NRF under test verifies that the iat value is valid.

Pre-Conditions:

- Test environment with a consumer NF and a SCP, which may be simulated. (Potentially simulated) consumer NF and (potentially simulated) SCP can be combined for the testing purpose.
- The NRF under test is preconfigured with the certificate of the consumer NF.
- The NRF under test is configured to require assertions for NF consumer authentication for at least one of its services.
- The NRF under test has implemented the client credentials assertion (CCA) authentication method as specified in TS 33.501 [10], clause 13.3.8.3.
- The tester has the private key of the consumer NF.
- The tester has access to the interface between the consumer NF and the NRF under test.

Execution Steps

1. The tester computes a client credentials assertion correctly, except that the timestamp (iat) value is in the future. The tester should make sure that the expiration time is greater than the timestamp (iat) value.
2. The tester includes the client credentials assertion in the service request sent from the consumer NF to the NRF under test via the SCP.
3. The tester captures the response sent back by the NRF under test.

Expected Results:

The NRF under test rejects the consumer NF's service request with a 403 Forbidden HTTP status code and sends back an error message according to the description TS 29.500 [21], clause 6.7.5.

Expected format of evidence:

Evidence suitable for the interface, e.g., evidence can be presented in the form of log messages or a packet trace. A packet trace should at least contain the messages sent between the NRF and the NF service consumer.

4.2.2.4 NF access token procedure

4.2.2.4.1 NF access token request verification failure handling within single PLMN

Requirement Name: Access token request verification

Requirement Reference: TS 33.501 [3], clause 13.4.1.1.2

Requirement Description:

According to TS 33.501 [3], clause 13.4.1.1.2, the NRF verifies the input parameters in the access token request as follows:

- It checks that the NF Instance ID and, if available, NF type and PLMN ID(s), in the access token request match the corresponding ones in the public key certificate of the NF service consumer or those in the NF profile of the NF service consumer. If the verification of the access token request fails at the NRF, the NRF shall return "400 Bad Request" status code, and "error" attribute set to "invalid_client", as stated in TS 29.510 [5], clause 5.4.2.2, and the access token request is not further processed.
- Based on the input attributes (e.g., NF Type, PLMN ID(s), scope), it checks that the NF service consumer is authorized to access the requested services from the NF service producer. When certain parameters in the access token request are not supported by the NRF, the NRF shall ignore the unsupported parameters and continue processing the request with the rest of the parameters.

- If available, it verifies the S-NSSAIs of the NF service consumer and checks whether there are restrictions on the NF service consumer to access NF Service Producers' services of a specific NF type depending on the slices for which they offer their services.

If the access token request fails, the NRF shall return an appropriate error status code (e.g., "400 Bad Request", "401 Unauthorized", "403 Forbidden")

Threat References: TR 33.926 [6], clause H.2.2.Y, Elevation of privilege via failure to verify access token request

Test Case:

Test Name: TC_ACCESS_TOKEN_REQUEST_VERIFICATION_NRF

Purpose:

Verify that the NRF under test correctly handles access token request verification.

Pre-Conditions:

- The NRF under test is preconfigured with either the public key certificate or NF profile of the NF service consumer.
- The NRF under test is preconfigured with notional access token request authorization policies for an NF service consumer and the tester is provided documentation necessary to generate authorized access token requests.
- Test environment includes an NF service consumer.
- The NF service consumer may be simulated.

Execution Steps

Test Cases A~B are tests on failure handling by the NRF under test when the access token request failed verification

Test Case A: invalid client

1) The tester constructs an access token request correctly, except that the NF instance ID is different from the NF instance ID of the NF service consumer public key certificate. The NF service consumer sends the access token request to the NRF under test.
2) The NRF under test recognizes the NF instance ID does not match the NF instance ID of the NF service consumer public key certificate.
3) The access token request verification fails, and the NRF sends an appropriate response, as specified in TS 29.510 [5] clause 5.4.2.2 (e.g., 400 bad request, 403 forbidden, or 3XX for redirection).

Test Case B: unauthorized request

1) The tester constructs an access token request correctly, except that the input attributes (e.g., NF Type, PLMN ID(s), scope) are not authorized based on preconfigured authorization policies. The NF service consumer sends the access token request to the NRF under test.
2) The NRF under test recognizes that at least one of the input attributes implies the NF service consumer is not authorized for the requested services
3) The access token request verification fails, and the NRF sends an appropriate response, as specified in TS 29.510 [5] clause 5.4.2.2 (e.g., 400 bad request, 403 forbidden, or 3XX for redirection).

Expected Results:

For test cases A~B on verification failure of the access token request parameters, the NRF under test rejects the NF service consumer’s service request.

Expected format of evidence:

Evidence suitable for the interface, e.g., evidence can be presented in the form of log messages or a packet trace. A packet trace should at least contain the messages sent between the NRF and the NF service consumer.

4.2.3 Technical baseline

4.2.3.1 Introduction

The present clause provides baseline technical requirements.

4.2.3.2 Protecting data and information

4.2.3.2.1 Protecting data and information – general

There are no NRF-specific additions to clause 4.2.3.2.1 of TS 33.117 [2].

4.2.3.2.2 Protecting data and information – unauthorized viewing

There are no NRF-specific additions to clause 4.2.3.2.2 of TS 33.117 [2].

4.2.3.2.3 Protecting data and information in storage

There are no NRF-specific additions to clause 4.2.3.2.3 of TS 33.117 [2].

4.2.3.2.4 Protecting data and information in transfer

There are no NRF-specific additions to clause 4.2.3.2.4 of TS 33.117 [2].

4.2.3.2.5 Logging access to personal data

There are no NRF-specific additions to clause 4.2.3.2.5 of TS 33.117 [2].

4.2.3.3 Protecting availability and integrity

There are no NRF-specific additions to clause 4.2.3.3 of TS 33.117 [2].

4.2.3.4 Authentication and authorization

There are no NRF-specific additions to clause 4.2.3.4 of TS 33.117 [2].

4.2.3.5 Protecting sessions

There are no NRF-specific additions to clause 4.2.3.5 of TS 33.117 [2].

4.2.3.6 Logging

There are no NRF-specific additions to clause 4.2.3.6 of TS 33.117 [2].

4.2.4 Operating Systems

There are no NRF-specific additions to clause 4.2.4 of TS 33.117 [2].

4.2.5 Web Servers

There are no NRF-specific additions to clause 4.2.5 of TS 33.117 [2].

4.2.6 Network Devices

There are no NRF-specific additions to clause 4.2.6 of TS 33.117 [2].

4.3 NRF-specific adaptations of hardening requirements and related test cases

4.3.1 Introduction

The requirements proposed hereafter (with the relative test cases) aim to securing NRF by reducing its surface of vulnerability. In particular, the identified requirements aim to ensure that all the default configurations of NRF (including operating system software, firmware and applications) are appropriately set.

4.3.2 Technical baseline

All text from TS 33.117, clause 4.3.2 also applies to NRFs. There are no NRF-specific adaptations or additions to clause 4.3.2 of TS 33.117 [2].

4.3.3 Operating systems

There are no NRF-specific additions to clause 4.3.3 of TS 33.117 [2].

4.3.4 Web servers

There are no NRF-specific additions to clause 4.3.4 of TS 33.117 [2].

4.3.5 Network devices

There are no NRF-specific additions to clause 4.3.5 of TS 33.117 [2].

4.3.6 Network functions in service-based architecture

There are no NRF-specific additions to clause 4.3.6 in TS 33.117 [2].

4.4 NRF-specific adaptations of basic vulnerability testing requirements and related test cases

4.4.1 Introduction

There are no NRF specific addtions to clause 4.4.1 of TS 33.117 [2].

4.4.2 Port Scanning

There are no NRF specific addtions to clause 4.4.2 of TS 33.117 [2].

4.4.3 Vulnerability scanning

There are no NRF specific addtions to clause 4.4.3 of TS 33.117 [2].

4.4.4 Robustness and fuzz testing

The test cases under clause 4.4.4 of TS 33.117 [2] are applicable to NRF.

The interface defined for the NRF are in 4.2.3 of TS 23.501 [7].

According to clause 4.4.4 of TS 33.117 [2], the transport protocols available on the interfaces providing IP-based protocols need to be robustness tested. Following TCP/IP layer model and considering all the protocols over transport layer, for NRF, the following interface and protocols are in the scope of the testing:

- For Nnrf: the TCP, HTTP2 protocols and the format of JSON.

NOTE: There could be other interfaces and/or protocols requiring testing under clause 4.4.4 of TS 33.117 [2]

Annex A (informative): Change history

Change history							
date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2019-09	SA#85					Change control version	16.0.0
2019-10						EditHelp review	16.0.1
2019-12	SA#86	SP-191138	0001	-	F	Adding abbreviations and corrections for alignment	16.1.0
2020-07	SA#88E	SP-200358	0002	1	F	Update to the test case of NF discovery authorization for specific slice	16.2.0
2022-03	-	-	-	-	-	Update to Rel-17 version (MCC)	17.0.0
2023-06	SA#100	SP-230677	0003	1	B	Robustness interfaces and protocols defined for NRF	18.0.0
2023-06	SA#100	SP-230677	0004	-	F	SCAS release reference corrections	18.0.0
2025-07	SA#108	SP-250657	0008	1	F	NF discovery authorization based on expected NF profile	19.0.0
2026-03	SA#111	SP-260161	0011	-	B	Add new test cases on Indirect Communication and token verification	20.0.0